

Personal Finance Advisor — Project Brief & Build Roadmap

Purpose: This PDF is meant to preserve the full project context so a fresh chat can guide the build from scratch without losing direction.

1) Project Title

Personal Finance Advisor

A personal expense tracking and financial insight web app that begins with expense logging and category-wise dashboard analytics, and later grows into a machine-learning powered finance assistant.

2) Core Project Idea

The user will add expenses manually, view their spending behavior by category on a dashboard, and later receive ML-based predictions such as next month spending, budget risk, and simple automated financial advice.

3) Dataset Currently Available

A synthetic finance dataset with **1000 records** has already been prepared and downloaded for the project.

Dataset Columns:

- user_id
- expense_date
- amount
- category
- payment_method
- note

Category Values:

- Food
- Groceries
- Rent
- Utilities
- Transport
- Shopping
- Entertainment
- Health
- Education
- Travel
- EMI
- Subscriptions

- Miscellaneous

This dataset is enough to start the first build phase: storing expenses and showing category-wise analytics. It will also serve as the starting data source for future feature engineering when ML models are added.

4) Important Technical Decision

The first version of the app does **not** need a machine learning model. Expense categorization and dashboard analytics will be done using normal database queries and aggregation logic (for example, grouping expenses by category and summing amounts). ML will only be introduced later for prediction tasks.

5) Features to Build

- **Phase 1 Features (No ML):** Add expense, store expense, show all expenses, show category-wise spending, monthly total, dashboard charts.
- **Phase 2 Features (First ML):** Predict next month total spending.
- **Phase 3 Features:** Budget risk / overspending prediction.
- **Phase 4 Features (Optional):** Spending habit insights, automated advice, simple user behavior analysis.

6) Recommended Tech Stack

- **Frontend:** React
- **Styling:** CSS (or Tailwind later if desired)
- **Backend:** Flask or FastAPI (recommended because ML integration will be easier later)
- **Database:** MongoDB or MySQL
- **Machine Learning:** Python, Pandas, scikit-learn
- **Visualization:** Recharts / Chart.js on frontend

7) Phase-Wise Roadmap

Phase 1 — Expense Tracker + Dashboard (No ML)

- Build expense entry form
- Store expenses in database
- Display all user expenses
- Build dashboard cards: total spend, total transactions
- Show category-wise spending using bar chart / pie chart
- Filter by month if possible

Phase 2 — First ML Model: Next Month Spending Prediction

- Prepare historical monthly spending data
- Engineer features from expense history
- Train first simple model to predict next month total spending
- Integrate prediction result into dashboard

Phase 3 — Budget Risk Prediction

- Define budget threshold / monthly budget

- Train classification model for overspending risk
- Show user if they are likely to exceed budget

Phase 4 — Advice & Smart Insights (Optional)

- Add simple financial advice rules
- Identify risky spending categories
- Optional: add spending behavior clustering or smarter advice later

8) Build Philosophy for This Project

This project should be built in a clean progression: first data collection and dashboarding, then ML. The aim is not to use ML unnecessarily. Instead, ML should only be used where prediction or intelligent decision-making is actually needed.

9) What a New Chat Should Know Immediately

- This is a Personal Finance Advisor project.
- The user wants to build it from scratch and focus strongly on the ML learning side.
- The project should start simple and be built in phases.
- Current focus is Phase 1: expense entry + category dashboard.
- No ML model is needed for Phase 1.
- A 1000-record dataset is already available with 6 columns and fixed finance categories.

This document can be uploaded or referenced in a fresh chat to continue the project build with full context.